

The Makings of A Hockey Champion

Using Data to Analyze Key Features in the NHL



National Hockey League Playoffs

- Playoffs lend themselves to potential upsets due to small occurrences.
- Can data help us narrow in on who is truly "the best" outside of the playoffs?
- Can we determine what details help predict a winning team?



Ethical Concerns



No personal, confidential information is used in this dataset. All information is freely available to anyone watching games.



Sports betting can be a fun pastime, but also a legitimate concern for those affected by gambling addictions. Results of this report should not be taken as a guarantee of betting income.



Scope of Data

- Games from 2000-2020
- Regular season and Playoff seasons
- Individual player stats, overall team stats
- Individual game scores



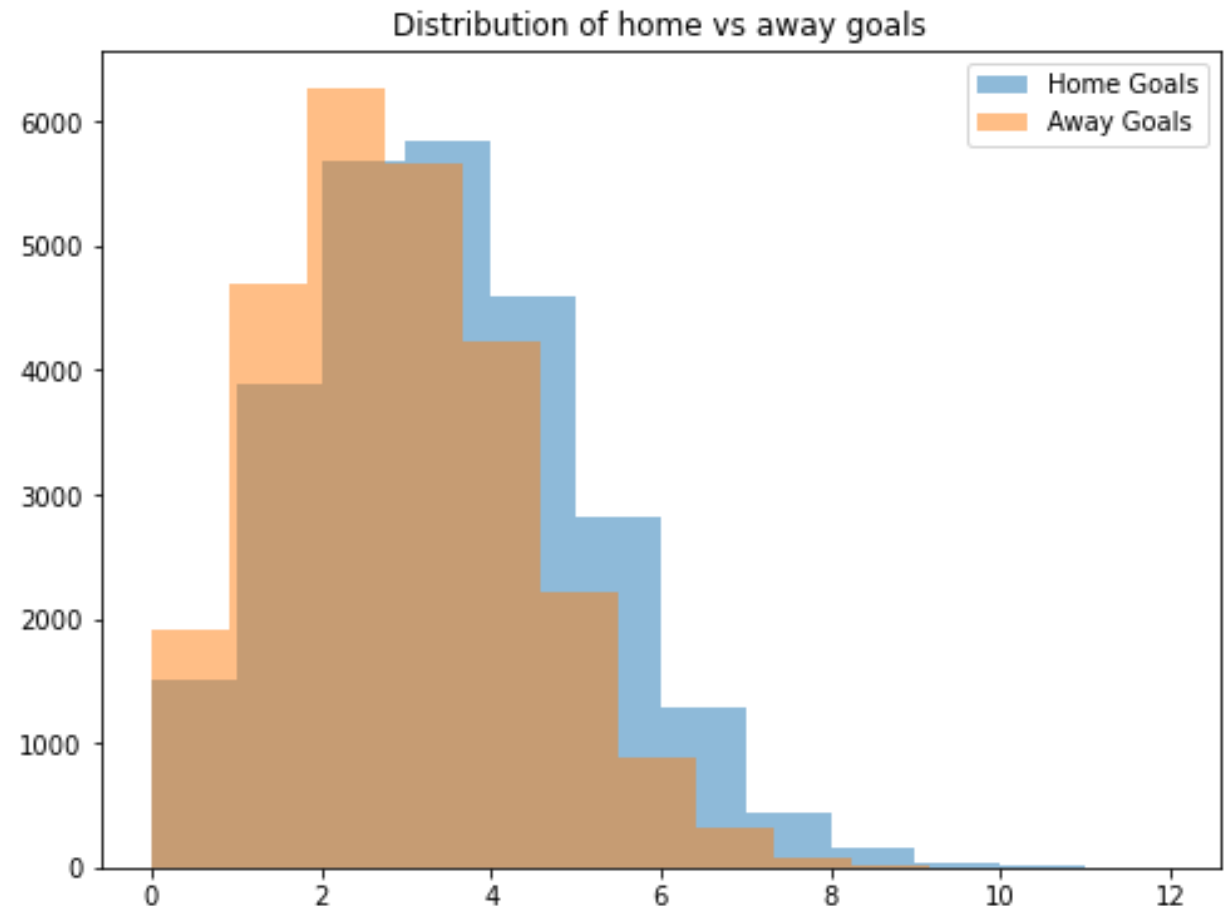
Metric: Goals per game

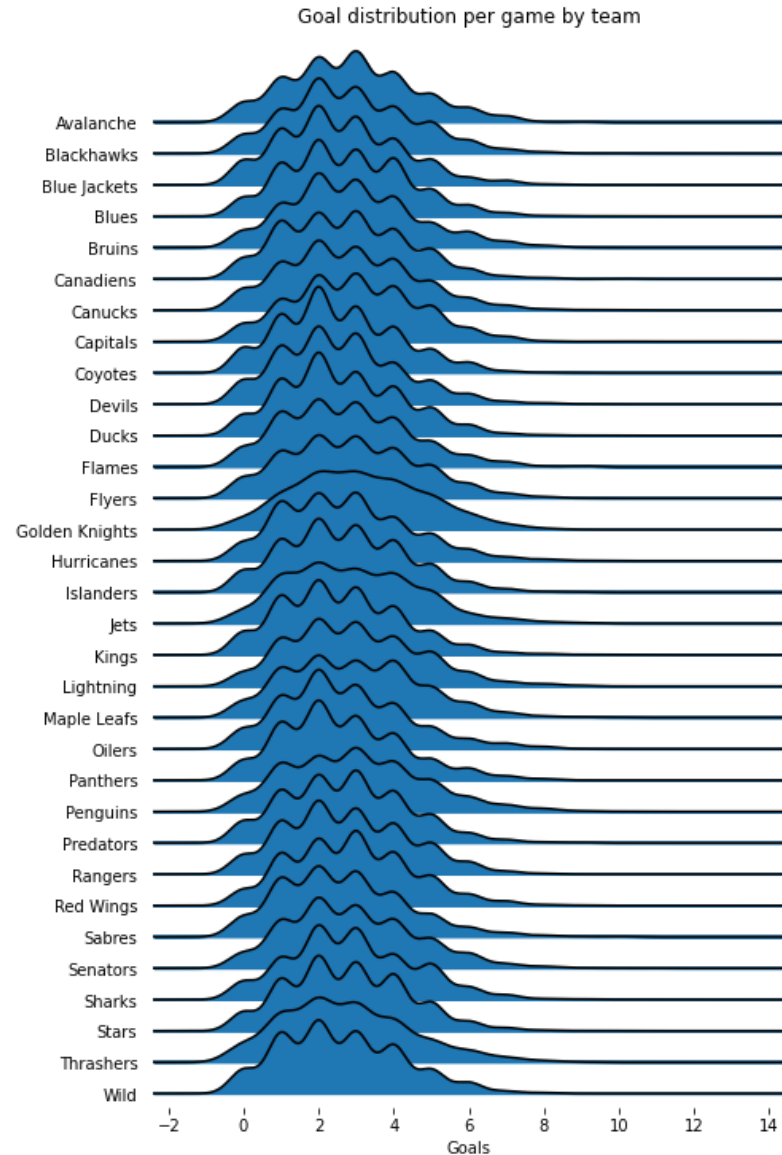
- Hockey is a low-scoring game, so the distribution of goals made may show a significant difference amongst teams or players, regardless of games won or lost



Home Team Advantage

- Many fans swear by the advantage a team has when playing at their home arena. Visually, we do not see much of a difference in goals scored per team.
- The distributions look similar, but a Kolmogorov-Smirnov 2-sample test for goodness of fit gives a p-value of $6.01e-53$, which is below the 0.05 threshold of statistical significance. This means that the fans are right, games played in a home arena tend to score more goals.

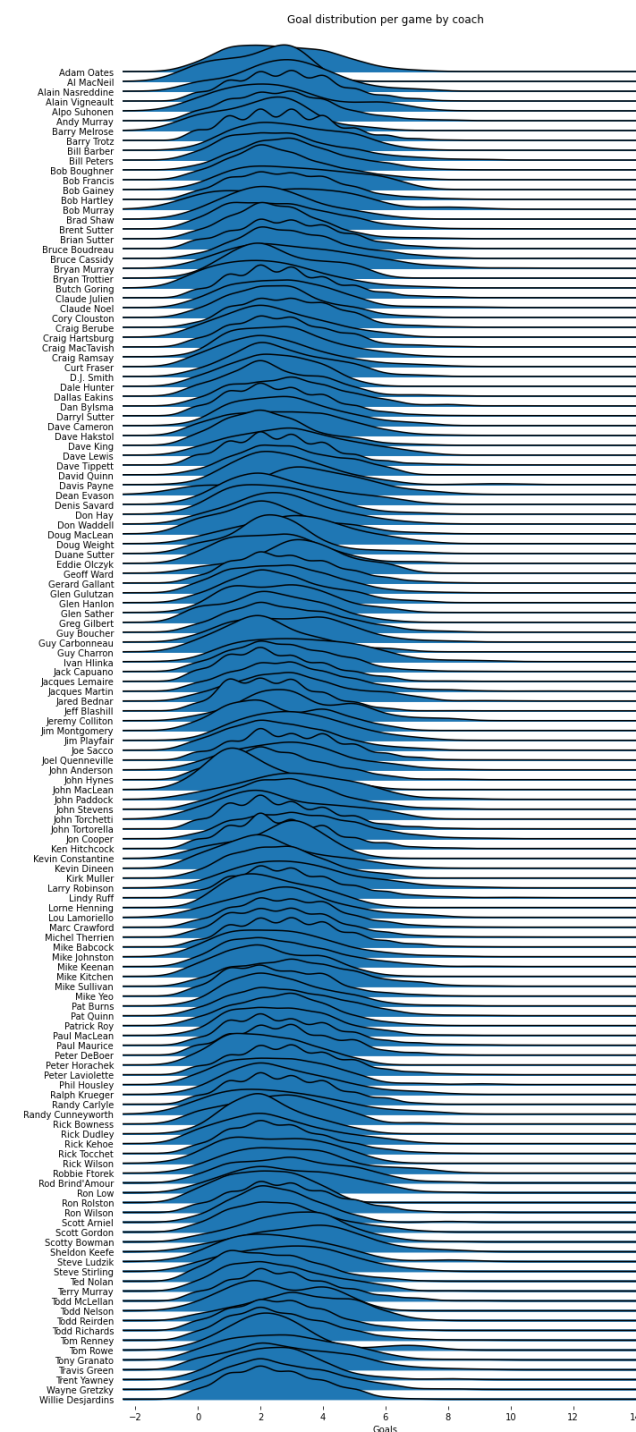




- Looking at team goal distribution for the breadth of data, there are no distinct outliers to show a team who tends to score more or less goals on average.

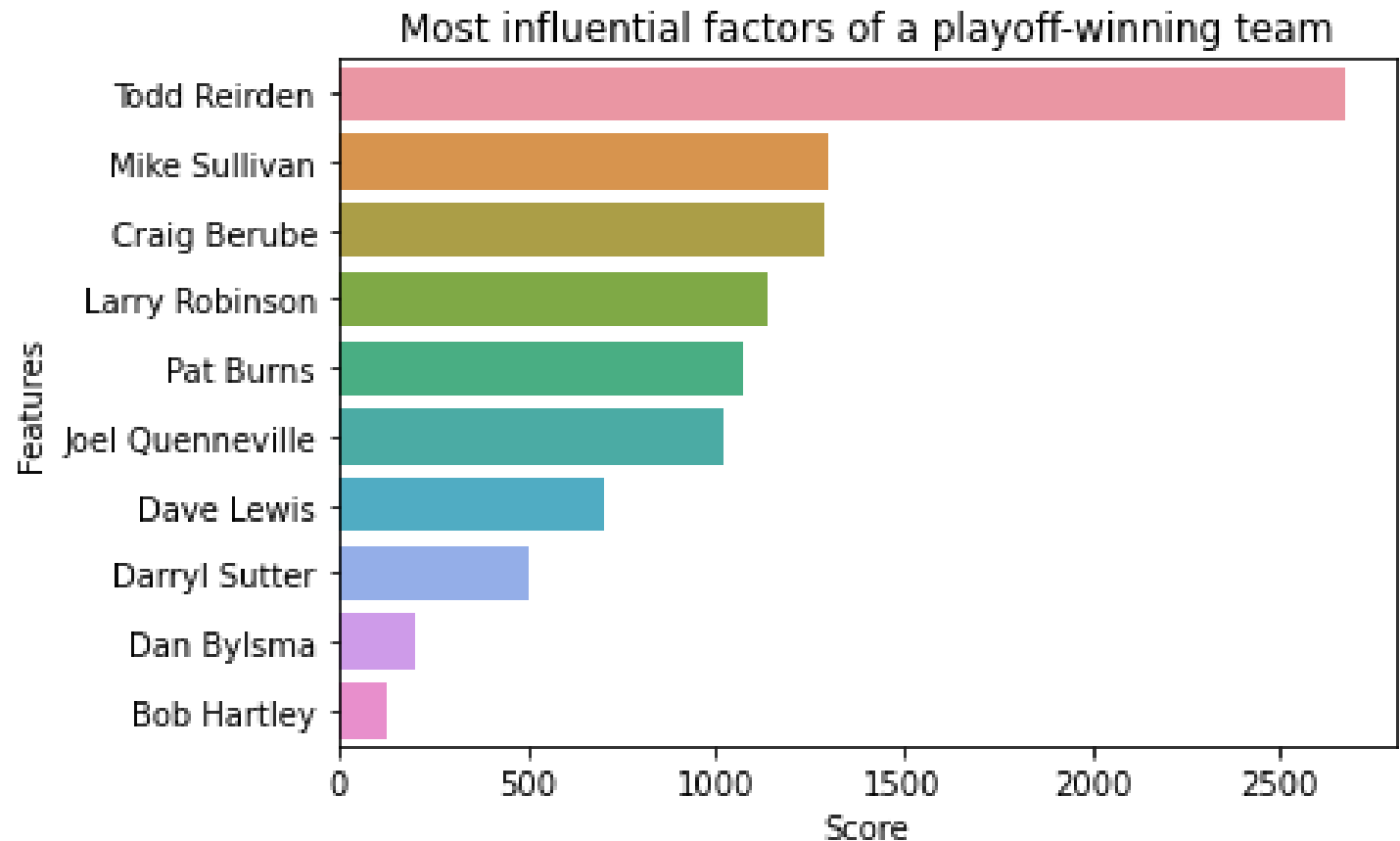


- This graph is crowded, but shows a similar comparison between different coaches. No singular coach tends to have a distinct advantage or disadvantage when it comes to total goals scored per game.



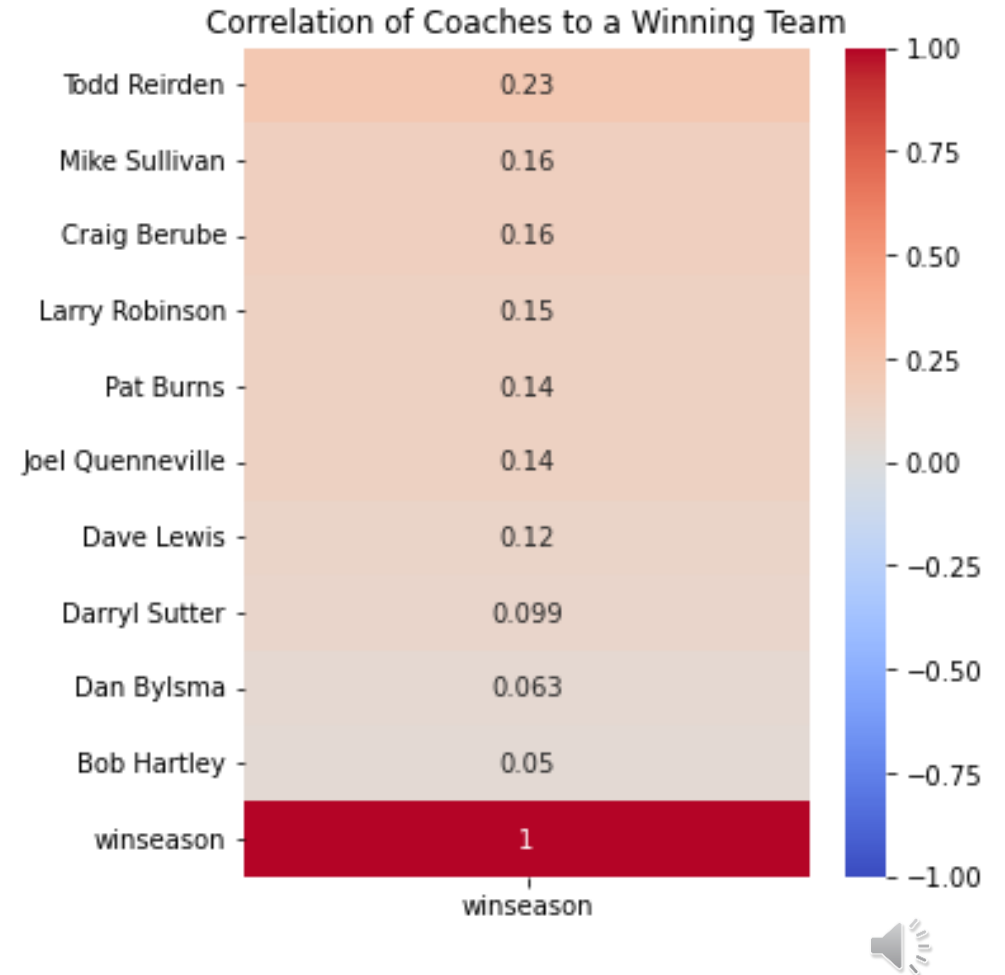
Metrics: Most Influential Factors

- Feature Selection Model
 - Determines features that have the strongest effect on the outcome (Playoff-winning team)



Interpreting winning features

- Selecting the most influential features, or coaches, is only helpful if we know what to do with this information.
- Looking at correlation scores, we can see that these coaches all have a positive correlation. Their presence on a team contributes to the overall chance that a team will win that season's playoffs.



Most Influential Factors



A full-on model will not provide the helpful information for building a future team.



Which factors have the strongest influence on determining a playoff winner?

People change, and there are always new players and coaches



Conclusions



Despite having data from 20 years, we don't see an obvious "best" team, either in goals or number of playoff wins.



Team managers should look to the list of coaches to determine good choices for future team recruits



Questions?

- Will this project be able to predict next year's playoff winner?
- Who should we be keeping an eye on for fantasy teams in the next season?
- Why bother creating this if it only has recommendations?
- How does this help a team in the recruiting process?
- Why not just look at playoff winners and bet on the recent ones?
- Is there money to be made from this project?
- Is any of this applicable to other sports?
- What can an aspiring professional player use from this project?
- How can there be precautions against excessive sports betting using this model?
- What data are you looking to add in the future to refine the model?

